1. A population consists of five numbers 3, 6, 9, 15 and 27. Consider all samples of size 3 that can be drawn without replacement from this population. Find i) The population mean ii) The population standard deviation iii) The mean of the sampling distribution of the means iv) The standard deviation of the sampling distribution of the means.

Sol: A population consists of five numbers 3, 6, 9, 15 and 27.

Given population is 3, 6, 9, 15, 27

Size of the population is $N = 5$

sample size is $n = 3$

(i) Mean of the population $= \mu = \frac{3+6+9+15+27}{5}$

$= \frac{60}{5}$

$= 12$

(ii) Standard deviation of the population

$$\sigma = \sqrt{\sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}}$$

x_i = ith element in the population

where $\mu = 12$

$$= \sqrt{\frac{(3-12)^2 + (6-12)^2 + (9-12)^2 + (15-12)^2 + (27-12)^2}{5}}$$

$$= \sqrt{\frac{81+36+9+9+225}{5}}$$

$$= \sqrt{\frac{360}{5}}$$

$$= \sqrt{72}$$

$$= 8.4853$$

(iii) When sample of size $n = 3$ is drawn from the population without replacement No of samples of size $n = 3$ that can be drawn from population of size $N = 5$ without replacement is given by $Nc_n = 5c_3 = 10$

Samples are (3,6,9) (3,6,15) (3,6,27) (3,9,15) (3,9,27)
(3,15,27) (6,9,15) (6,9,27) (6,15,27) (9,15,27)

Mean of the above samples are

6, 8, 12, 9, 13, 15, 10, 14, 16, 17

Now mean of sampling distribution of means is

$$\mu_{\bar{x}} = \frac{\text{Sum of sample means}}{\text{No of sample means}}$$

$$= \frac{6+8+12+9+13+15+10+14+16+17}{10}$$

$$= \frac{120}{10}$$

$$= 12$$

Hence $\boxed{\mu_{\bar{x}} = 12 = \mu}$

(iv) Standard deviation of sampling distribution of means

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$= \frac{8.4853}{\sqrt{3}} \sqrt{\frac{5-3}{5-1}}$$

$$= \frac{8.4853}{\sqrt{3}} \sqrt{\frac{2}{4}}$$

$$= \frac{8.4853}{\sqrt{3}} \times \frac{1}{\sqrt{2}}$$

$$= \frac{8.4853}{\sqrt{6}}$$

$$= 3.464$$

Final answers:

$\boxed{\mu = 12}$

$\boxed{\sigma = 8.4853}$

$\boxed{\mu_{\bar{x}} = 12}$

$\boxed{\sigma_{\bar{x}} = 3.464}$

2. A population consists of the numbers 2, 3, 6, 8 and 11. Consider all possible samples of size 2 that can be drawn with replacement from this population. Find i) The population mean ii) The population standard deviation iii) The mean of the sampling distribution of the means iv) The standard deviation of the sampling distribution of the means.

Given population = 2, 3, 6, 8, 11

Size of the population is $N=5$

sample size is $n=2$

(i) Mean of the population

$$\mu = \frac{2+3+6+8+11}{5}$$

$$= \frac{30}{5}$$

$$\boxed{\mu = 6}$$

(ii) Standard deviation of the population

$$\sigma = \sqrt{\sum_{i=1}^{N} \frac{(x_i-\mu)^2}{N}}$$

$$= \sqrt{\frac{(2-6)^2+(3-6)^2+(6-6)^2+(8-6)^2+(11-6)^2}{5}}$$

$$= \sqrt{\frac{16+9+0+4+25}{5}}$$

$$= \sqrt{\frac{54}{5}}$$

$$\sigma = \sqrt{10.8} = 3.286$$

(iii) Mean of the sampling distribution of mean

Since with replacement:

No. of samples $= N^n = 5^2 = 25$

Samples are: (2,2), (2,3), (2,6), (2,8), (2,11)

(3,2), (3,3), (3,6), (3,8), (3,11)

(6,2), (6,3), (6,6), (6,8), (6,11)

(8,2), (8,3), (8,6), (8,8), (8,11)

(11,2), (11,3), (11,6), (11,8), (11,11)

Their sample means are: 2, 2.5, 4, 5, 6.5, 2.5, 3, 4.5, 5.5, 7, 4, 4.5, 6, 7, 8.5, 5, 5.5, 7, 8, 9.5, 6.5, 7, 8.5, 9.5, 11

$$\mu_{\bar{x}} = \frac{\text{Sum of sample means}}{\text{No of samples}}$$

$$= \frac{150}{25}$$

$$\mu_{\bar{x}} = 6 = \mu$$

(ii) Standard deviation of sampling distribution of mean

$$\sigma_{\bar{x}} = \sqrt{\frac{\Sigma(\bar{x}_i - \mu_{\bar{x}})^2}{25}}$$

$$= \sqrt{\frac{16+12.25+4+1+0.25+12.25+9+2.25+0.25+1+4+2.25+0+1+6.25+1+0.25+1+4+12.25+0.25+1+6.25+12.25+25}{25}}$$

$$= \sqrt{\frac{116.25}{25}}$$

$$= \sqrt{4.65}$$

$$\sigma_{\bar{x}} \simeq 2.156$$

Final Answers

$\mu = 6$

$\sigma = 3.286$

$\mu_{\bar{x}} = 6$

$\sigma_{\bar{x}} = 2.156$

3. Find the median for the following table

Group	50-54	55-59	60-64	65-69	70-74	75-79
Frequency	4	8	10	9	7	6

Soln

Step1: Find cumulative Frequency

Group	f	Cumulative Frequency
50-54	4	4
55-59	8	4+8=12
60-64	10	12+10=22
65-69	9	22+9=31
70-74	7	31+7=38
75-79	6	38+6=44
	N=44	

Step2: Find Median Class

$\frac{N}{2} = \frac{44}{2} = 22$

The cumulative frequency equal to 22 occurs in the class 60-64

Median class = 60-64

Step3: Identify values

- $L = 59.5$ (lower class boundary of median class)
- $f = 10$
- $C = 12$ (cumulative frequency before medianclass)
- $h = 5$
- $N = 44$

Step 4: Apply Formula

$$\text{Median} = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$

$$= 59.5 + \frac{5}{10}(22-12)$$

$$= 59.5 + \frac{5}{10}(10)$$

$$= 59.5 + 5$$

$$= 64.5$$

Median = 64.5

4. What are the types of data? Explain the methods of collection of data.

Data: Data is a tool that helps an investigator in understanding a problem by providing the required information. Data is classified into two types: Primary Data and secondary data.

Types of Data

1. Primary Data:

Primary data is the data originally collected by an investigator for the first time for any statistical investigation. It is collected directly from the original source.

Examples: surveys, interviews, questionnaires, direct observation.

2. Secondary Data:

Secondary data is the data has already been collected by some person or agency for statistical investigation.

Examples: Books, journals, government reports, websites, newspapers.

Methods of collection of Data

Methods of collecting Primary Data:

Direct Personal Investigation

Indirect Personal Investigation

Information from Local correspondents

Information through questionnaires and schedules

Mailing Method

Enumerator's method

Methods of collecting secondary data:

Government Publications

Semi-Government Publications

Publications of Trade Associations

Journals and Papers

International Publications

Publications of Research Institutes

Unpublished sources

Conclusion:

Thus, data is of two types - Primary Data and secondary Data - and can be collected through various primary and secondary methods depending on the purpose of the investigation.

5. A random sample of size 100 is taken from an infinite population having the mean $\mu = 76$ and variance $\sigma^2 = 256$. Find the probability that will be i) lies between 75 and 78 ii) More than 72.

Given

Population mean (μ) = 76

Population variance (σ^2) = 256

sample size (n) = 100

So, $\sigma = \sqrt{256} = 16$

Standard error (SE) $= \frac{\sigma}{\sqrt{n}} = \frac{16}{\sqrt{100}} = \frac{16}{10} = 1.6$

Formula:

$$Z = \frac{\bar{X} - \mu}{\sigma}$$

$$P(a < \bar{x} < b) = P(z_1 < z < z_2)$$

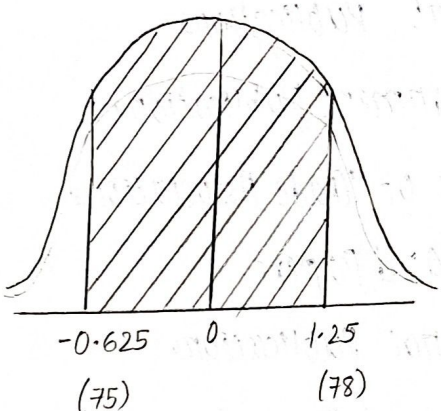
$$P(\bar{X} > a) = P(z > z_a)$$

(i) Probability that the sample mean lies between 75 and 78.

$P(75 < \bar{X} < 78)$

For 75: $Z_1 = \frac{75-76}{1.6} = \frac{-1}{1.6} = -0.625$

For 78: $Z_2 = \frac{78-76}{1.6} = \frac{2}{1.6} = 1.25$



From z table:

$P(0 < Z < 1.25) = 0.3944$

$P(0 < Z < 0.625) = 0.2340$

Therefore,

$P(-0.625 < Z < 1.25) = 0.3944 + 0.2340$

$= 0.6284$

$\therefore \boxed{P(75 < \bar{X} < 78) = 0.6284}$

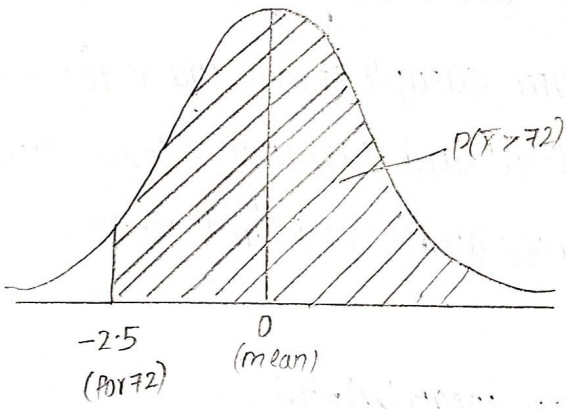
(ii) Probability that the sample mean is more than 72.

$P(\bar{X} > 72)$

For 72: $Z = \frac{72-76}{1.6}$

$= \frac{-4}{1.6}$

$= -2.5$



From z-table

$P(0 < Z < 2.5) = 0.4938$

So, $P(Z > -2.5) = 0.5 + 0.4938 = 0.9938$

$\boxed{P(\bar{X} > 72) = 0.9938}$

6. Find 95% confidence limits for the mean of a normally distributed population from which the following sample was taken 15, 17, 10, 18, 16, 9, 7, 11, 13, 14, 20, 13, 19, 25, 18, 16.

Soln Given $n = 16$

Sample: 15, 17, 10, 18, 16, 9, 7, 11, 13, 14, 20, 13, 19, 25, 18, 16

1) Sample mean ($\bar{x}$):

$$\bar{x} = \frac{\sum x}{n} = \frac{15+17+10+18+16+9+7+11+13+14+20+13+19+25+18+16}{16}$$

$$\bar{x} = \frac{241}{16} = 15.0625$$

2) Sample variance (s^2):

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{15} \sum (x_i - 15.0625)^2$$

$$s^2 = 20.996$$

3) Sample standard deviation (s):

$$s = \sqrt{s^2} = \sqrt{20.996} = 4.582$$

4) Degrees of freedom (v):

$$v = n - 1 = 16 - 1 = 15$$

5) t-value:

For 95% confidence $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \Rightarrow \frac{\alpha}{2} = 0.025$

For $v = 15$, $t_{0.025} = 2.131$

6) 95% confidence interval:

$$\bar{x} - t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

$$15.0625 - 2.131 \frac{4.582}{\sqrt{16}} < \mu < 15.0625 + 2.131 \frac{4.582}{\sqrt{16}}$$

$$15.0625 - 2.441 < \mu < 15.0625 + 2.441$$

$$12.62 < \mu < 17.50$$

95% confidence limits for the population mean are 12.62 and 17.50

7. Two researchers adopted different sampling techniques while investigating the same group of students to find the number of students falling in different intelligence levels.

The results are as follows:

Researcher	No. of students in each level				Total
	Below Average	Average	Above Average	Genius	
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

Would you say that the sampling techniques adopted by two researchers are significantly different?

Step 1: State the Hypothesis (H_0):

H_0: There is no significant difference between the sampling techniques adopted by the two researchers.

Step 2: Level of significance:

Assume level of significance $\alpha = 0.05$

Step 3: Test statistic:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Step 4: Find the expected frequencies:

$$E_i = \frac{\text{Row total} \times \text{column total}}{\text{Grand total}}$$

(a) Observed frequencies:

Researcher	No. of students in each level				
	Below Average	Average	Above Average	Genius	Total
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

(b) Expected frequencies:

Researcher	No. of students in each level				
	Below Average	Average	Above Average	Genius	Total
X	84	62	46	8	200
Y	42	31	23	4	100
Total	126	93	69	12	300

(c) Calculation of χ^2:

$\chi^2 = 0.0476 + 0.0645 + 0.0870 + 0.5000 + 0.0952 + 0.1290 + 0.1739 + 1.0000$

$= 2.0973 \approx 2.10$

Step 5: Conclusion:

Degrees of freedom: $df = (r-1)(c-1) = (2-1)(4-1) = 3$

At $\alpha = 0.05$, $\chi^2_{critical} = 7.815$ (from table)

Since $\chi^2 = 2.10 < 7.815$, we accept H_0.

Therefore, there is no significant difference between the sampling techniques adopted by the two researchers.

8. The mean yield of wheat from a district was 210 pounds with S.D 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D 12 pounds from a sample of 50 plots. Test whether there is any significant difference between the mean yields of crops in the two districts.

(b) Expected frequencies:

Researcher	No. of students in each level				
	Below Average	Average	Above Average	Genius	Total
X	84	62	46	8	200
Y	42	31	23	4	100
Total	126	93	69	12	300

(c) Calculation of χ^2:

$\chi^2 = 0.0476 + 0.0645 + 0.0870 + 0.5000 + 0.0952 + 0.1290 + 0.1739 + 1.0000$

$= 2.0973 \approx 2.10$

Step 5: Conclusion:

Degrees of freedom: $df = (r-1)(c-1) = (2-1)(4-1) = 3$

At $\alpha = 0.05$, $\chi^2_{critical} = 7.815$ (from table)

Since $\chi^2 = 2.10 < 7.815$, we accept H_0.

Therefore, there is no significant difference between the sampling techniques adopted by the two researchers.

8. The mean yield of wheat from a district was 210 pounds with S.D 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D 12 pounds from a sample of 50 plots. Test whether there is any significant difference between the mean yields of crops in the two districts.

Given: District 1: $\bar{X}_1 = 210$, $S_1 = 10$, $n_1 = 100$

District 2: $\bar{X}_2 = 220$, $S_2 = 12$, $n_2 = 50$

Since both n_1 and n_2 are ≥ 30, use z-test for difference between two means (large samples).

1. Hypotheses

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$ (Two-tailed test)

2. Level of significance

Since it is not given, take $\alpha = 0.05$

For a two-tailed test, $Z_{\frac{\alpha}{2}} = 1.96$

5% level of significance is generally used.

3. Test statistic

For two large samples,

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Substitute the values:

$$Z = \frac{210-220}{\sqrt{\frac{10^2}{100} + \frac{12^2}{50}}} = \frac{-10}{\sqrt{1+2.88}} = \frac{-10}{1.97} \approx -5.08$$

Therefore, $|z| = 5.08 > 1.96$

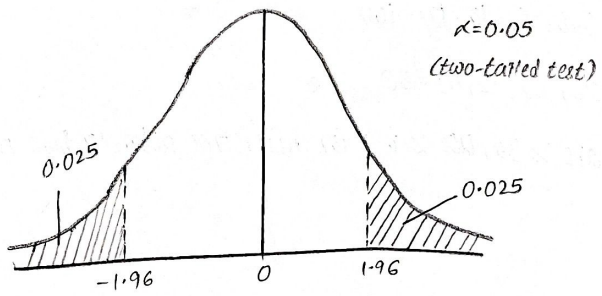
4. Decision

Since $|z| > Z_{\frac{\alpha}{2}}$

We reject H_0.

5. Conclusion

Hence, there is a significant difference between the mean yields of wheat in the two districts at 5% level of significance.



9. Ten soldiers participated in a shooting competition in the first week. After intensive training they participated in the competition in the second week. Their scores before and after training are given as follows:

Scores Before	67	24	57	55	63	54	56	68	33	43
Scores after	70	38	58	58	56	67	68	75	42	38

Do the data indicate that the soldiers have been benefited by the training?

Given

Ten soldiers participated in a shooting competition. Their scores before and after training are given below:

Before training	67	24	57	55	63	54	56	68	33	43
After training	70	38	58	58	56	67	68	75	42	38

$n = 10$ (use paired sample t-test)

1. Hypotheses

$H_0: \mu_1 = \mu_2$ (no significant difference)

$H_1: \mu_2 > \mu_1$ (training has improved the scores)

2. Calculate the differences and test statistics

Take d = After - Before

Difference (d): 3, 14, 1, 3, -7, 13, 12, 7, 9, -5

$\sum d = 50 \Rightarrow \bar{d} = \frac{50}{10} = 5$

$\sum d^2 = 732$

$s^2 = \frac{1}{n-1}\left[\sum d^2 - n(\bar{d})^2\right] = \frac{1}{9}\left[732 - 10(5)^2\right]$

$= \frac{1}{9}[732 - 250] = \frac{1}{9}[482] = 53.56$

$S = \sqrt{53.56} = 7.32$

3. Test statistic

$$t = \frac{\bar{d}}{\frac{s}{\sqrt{n}}}$$

$$= \frac{5}{\frac{7.32}{\sqrt{10}}}$$

$$= \frac{5}{2.31}$$

$$= 2.16$$

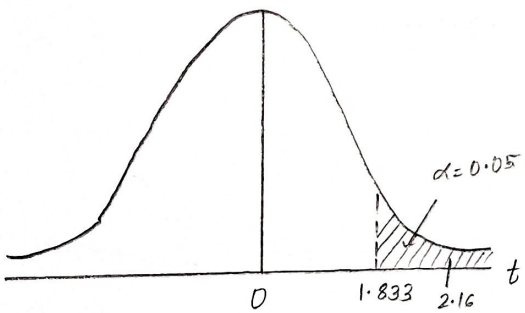
4. Critical value

At 5% level of significance, for 9 degrees of freedom, $t_{0.05} = 1.833$ (one-tailed test)

5. Decision

Since $2.16 > 1.833$, we reject H_0.

Graph (Right-tailed test)



Conclusion

The data indicate that the soldiers have been significantly benefited by the training at the 5% level of significance.

10. A sample of 64 students have a mean weight of 70 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs?

Given

$n = 64$

$\bar{x} = 70$ kgs

Population mean $(\mu) = 56$ kgs

Population standard deviation $(\sigma) = 25$ kgs

To test:

$H_0 : \mu = 56$

$H_1 : \mu \neq 56$ (Two tailed test)

Level of significance : $\alpha = 0.05$

Test statistic:

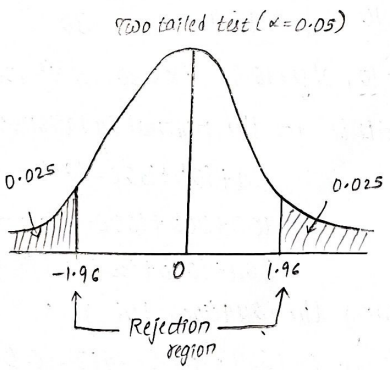
$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{70-56}{\frac{25}{\sqrt{64}}} = \frac{14}{\frac{25}{8}} = \frac{14 \times 8}{25} = \frac{112}{25} = 4.48$$

Critical value: For $\alpha = 0.05$ (two tailed), $Z_{0.025} = \pm 1.96$

Decision: Since $|z| = 4.48 > 1.96$

we reject H_0.

Conclusion: The sample is not from a population with mean weight 56 kgs (at 5% level of significance).



11. Fit a parabola to the following data

x	0	1	2	3	4
y	1	1·8	3·3	4·5	6·3

Assume a parabola of the form $y = a+bx+cx^2$

The normal equations are:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2 \longrightarrow (1)$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \longrightarrow (2)$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4 \longrightarrow (3)$$

Calculation table:

	x	y	x^2	xy	x^2y
	0	1	0	0	0
	1	1·8	1	1·8	1·8
	2	3·3	4	6·6	13·2
	3	4·5	9	13·5	40·5
	4	6·3	16	25·2	100·8
Σ	10	16·9	30	47·1	156·3

$\Sigma x = 10$, $\Sigma y = 16·9$, $\Sigma x^2 = 30$, $\Sigma x^3 = 100$, $\Sigma x^4 = 354$, $\Sigma xy = 47·1$, $\Sigma x^2y = 156·3$

Substitute in the normal equations:

$$5a + 10b + 30c = 16·9 \longrightarrow (1)$$

$$10a + 30b + 100c = 47·1 \longrightarrow (2)$$

$$30a + 100b + 354c = 156·3 \longrightarrow (3)$$

Solving the equations we get:

$$a = 0·9629,\ b = 0·8443,\ c = 0·1214$$

Hence the required parabola is:

$$\boxed{y = 0·9629 + 0·8443x + 0·1214x^2}$$

12. For the following data, fit the curve of the form $y = ae^{bx}$.

x	55	56	58	59	60	60	62
y	35	38	38	39	44	43	45

Given

$y = ae^{bx}$

Take logarithms on both sides:

$\ln y = \ln a + bx$ (Let $Y = \ln y$, $A = \ln a$)

Calculate the values of $Y = \ln y$

x	55	56	58	59	60	60	62
y	35	38	38	39	44	43	45
$Y = \ln y$	3.5553	3.6376	3.6376	3.6636	3.7842	3.7612	3.8067

Form the normal equations for $Y = A + bx$:

$$\sum Y = nA + b\sum x \longrightarrow (1)$$

$$\sum xY = A\sum x + b\sum x^2 \longrightarrow (2)$$

Calculate the required sums:

$\sum x = 421$, $\sum y = 26.8452$, $\sum x^2 = 26537$, $\sum xY = 1756.52$, $n = 7$

Substitute in (1) and (2)

$$26.8452 = 7A + 421b \longrightarrow (3)$$

$$1756.52 = 421A + 26537b \longrightarrow (4)$$

Solve the equations:

$$b = 0.0147, \quad A = 3.5434$$

Hence, the required exponential curve is

$$y = e^{3.5434} e^{0.0147x} = 34.58e^{0.0147x}$$

(approximately)

3. Calculate Rank correlation for the following data.

x	48	33	40	9	16	16	65	24	16	57
y	13	13	24	6	15	4	20	9	6	19

1. Assign ranks to x and y (in ascending order).

x	9	16	16	16	24	33	40	48	57	65
Rank (Rx)	1	2	3	4	5	6	7	8	9	10

y	4	6	6	9	9	13	13	15	19	24
Rank (Ry)	1	2	3	4	5	6	7	8	9	10

Find the difference in ranks ($d = Rx - Ry$) and square it.

No.	x	y	Rx	Ry	$d = Rx - Ry$	d^2
1	48	13	8	6	2	4
2	33	13	6	6	0	0
3	40	24	7	10	-3	9
4	9	6	1	2	-1	1
5	16	15	2	8	-6	36
6	16	4	3	1	2	4
7	65	20	10	9	1	1
8	24	9	5	4	1	1
9	16	6	4	2	2	4
10	57	19	9	9	0	0

$\sum d^2 = 4+0+9+1+36+4+1+1+4+0 = 60$

Use ~~Karl Pearson's~~ Spearman's rank correlation coefficient formula:

$$\gamma_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

Here $n = 10$ and $\sum d^2 = 60$

$$\gamma_s = 1 - \frac{6 \times 60}{10(10^2-1)} = 1 - \frac{360}{10(99)} = 1 - \frac{360}{990}$$

$$\gamma_s = 1 - 0.3636 = 0.6364$$

Hence, the rank correlation coefficient is

$$\gamma_s \simeq 0.636 \text{ (positive correlation)}$$

14. Calculate Karl Pearson's coefficient of correlation for the following data.

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

Sol: Use Karl Pearson's correlation coefficient formula:

$$\gamma = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x^2)][n\sum y^2 - (\sum y^2)]}}$$

Calculate the required sums from the data:

x	y	x^2	y^2	xy
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90

	11	11	121	121	121
	13	12	169	144	156
	7	3	49	9	21
Σ	70	63	728	651	676

$n=7,\ \Sigma x=70,\ \Sigma y=63,\ \Sigma x^2=728,\ \Sigma y^2=651,\ \Sigma xy=676$

Substitute the values in the formula:

$$r=\frac{7(676)-(70)(63)}{\sqrt{[7(728)-(70)^2]\,[7(651)-(63)^2]}}$$

$$r=\frac{4732-4410}{\sqrt{[5096-4900][4557-3969]}}$$

$$r=\frac{322}{\sqrt{[196][588]}}=\frac{322}{339.25}=0.948$$

Hence, Karl Pearson's coefficient of correlation is

$$\boxed{r\approx 0.948 \text{ (positive correlation)}}$$

15. The equations of two regression lines are $7x-16y+9=0$ and $5y-4x-3=0$ find the coefficient of correlation and the means of x and y.

Write the regression lines in the form $y=a_1+b_1x$ and $x=a_2+b_2y$.

Given:

$$7x-16y+9=0 \Longrightarrow y=\frac{7}{16}x+\frac{9}{16} \longrightarrow (1)$$

$$5y-4x-3=0 \Longrightarrow x=\frac{5}{4}y+\frac{3}{4} \longrightarrow (2)$$

Identify the regression coefficients.

From (1) : $b_{yx}=\frac{7}{16}=0.4375$

From (2) : $b_{xy}=\frac{5}{4}=1.25$

Find the coefficient of correlation

$$r = \pm\sqrt{b_{yx} \cdot b_{xy}}$$

$$= \pm\sqrt{0.4375 \times 1.25}$$

$$= \pm\sqrt{0.546875}$$

$r = +0.739$ (both slopes are positive, so r is positive)

Find the means of x and y using the formula

$\bar{x} = (a_2 + b_2\bar{y})$ and $\bar{y} = (a_1 + b_1\bar{x})$

Here $a_1 = \frac{9}{16}$, $b_1 = \frac{7}{16}$, $a_2 = \frac{3}{4}$, $b_2 = \frac{5}{4}$

substitute and solve: $\bar{x} = \frac{3}{4} + \frac{5}{4}\bar{y} \rightarrow ③$

$$\bar{y} = \frac{9}{16} + \frac{7}{16}\bar{x} \rightarrow ④$$

From ③ and ④

$$\bar{x} = \frac{3}{4} + \frac{5}{4}\left(\frac{9}{16} + \frac{7}{16}\bar{x}\right) \Rightarrow \bar{x} = \frac{3}{4} + \frac{45}{64} + \frac{35}{64}\bar{x}$$

$$\bar{x} = \frac{35}{64}\bar{x} = \frac{93}{64} \Rightarrow \frac{29}{64}\bar{x} = \frac{93}{64} \Rightarrow \bar{x} = \frac{93}{29} = 3.21$$

$$\bar{y} = \frac{9}{16} + \frac{7}{16}(3.21) = \frac{9}{16} + \frac{2247}{256} = \frac{1383}{256} = 5.40$$

Hence $\boxed{r \simeq 0.739,\ \bar{x} \simeq 3.21,\ \bar{y} \simeq 5.40}$

16. Ten persons are assessed by three judges in a beauty contest. Determine the closeness between opinions of judges by rank correlation coefficient.

Score assigned by J-1	65	70	72	76	77	79	72	80	77	72
Score assigned by J-2	59	71	77	71	69	79	69	65	67	80

Arrange the scores in ranks (highest score = rank 1)

Person	1	2	3	4	5	6	7	8	9	10
Judge 1 (score)	65	70	72	76	77	79	72	80	77	72
Judge 1 (Rank)	10	7	6	5	4	1	6	2	4	6
Judge 2 (score)	59	71	77	71	69	79	69	65	67	80
Judge 2 (Rank)	10	5	2	5	7	1	7	9	8	3

Calculate the difference of ranks (d) and d^2

Person	1	2	3	4	5	6	7	8	9	10
R_1 (Judge 1)	10	7	6	5	4	1	6	2	4	6
R_2 (Judge 2)	10	5	2	5	7	1	7	9	8	3
$d = R_1 - R_2$	0	2	4	0	-3	0	-1	-7	-4	3
d^2	0	4	16	0	9	0	-1	49	16	9

$\sum d^2 = 104$

Substitute in Spearman's rank correlation coefficient formula.

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

Here $n = 10$, $\sum d^2 = 104$

$$r_s = 1 - \frac{6 \times 104}{10(10^2-1)}$$

$$= 1 - \frac{624}{10(99)}$$

$$= 1 - \frac{624}{990}$$

$$r_s = 1 - 0.6303 = 0.3697 \approx 0.37$$

Hence, the closeness between the opinions of the two judges is

$r_s \approx 0.37$ (positive correlation)